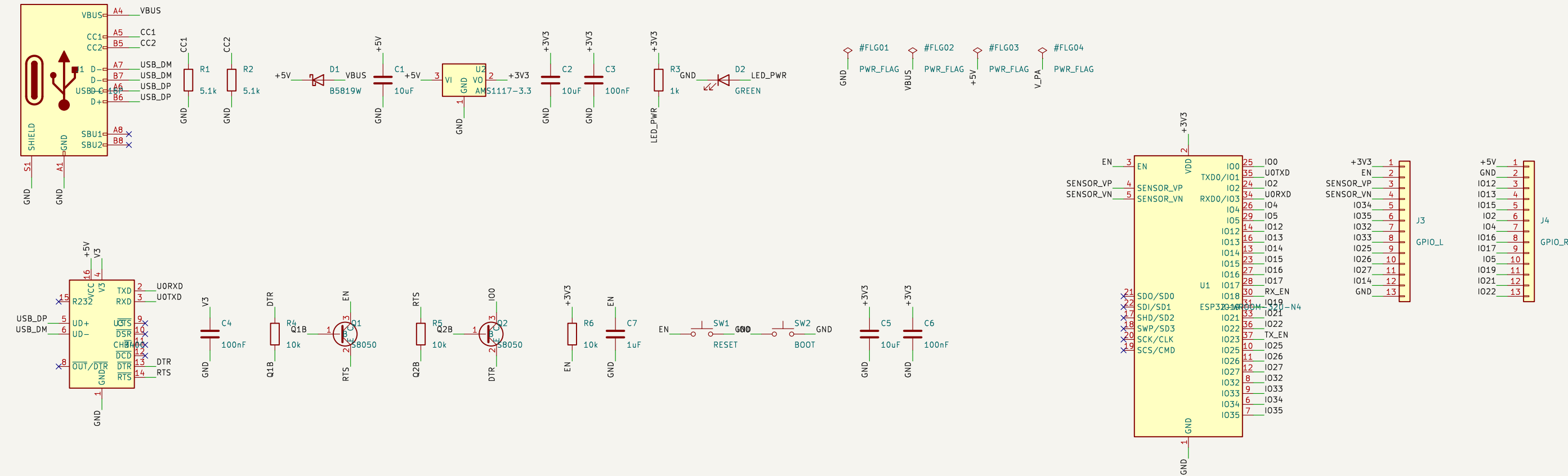


ESP32 Wi-Fi front-end board, rev A. Labels-only schematic: same net name = same net.
Power: USB-C 5V -> D1 -> AMS1117 3.3V. FEM supply is filtered through FB1 (+3V3_RF).
USB-serial: CH340C with DTR/RTS auto-reset (Q1/Q2), same as ESP32-DevKitC.



RF: ESP32-WROOM-32U U.FL -> short U.FL-U.FL pigtail -> J2 -> C8 -> pad R8/R9/R10 (OR / DNP)
-> RFX2401C (PA 20 dBm, LNA 12 dB) -> C12 -> pi filter L1/C13/C14 -> J5 SMA edge.
TX/RX switching: GPIO23 -> TXEN, GPIO18 -> RXEN via esp_wifi_set_ant_gpio() (see firmware/).
Keep ESP32 TX power at ~+8 dBm max with OR pad; RFX2401C input P1dB is about +5 dBm.

